Predictive Classification using Random Forest Ensembles

Technical Report - AIML Minor Project

Breast Cancer Diagnosis (Malignant vs Benign) on the Breast Cancer Wisconsin (Diagnostic) Dataset

Course / Semester	AIML, V Semester B.E.
Dataset	Kaggle: uciml/breast-cancer-wisconsin-data
Algorithm	RandomForestClassifier (scikit-learn)

Abstract. This project applies the Random Forest ensemble to the Breast Cancer Wisconsin (Diagnostic) dataset to classify tumours as malignant or benign from 30 numeric cell-nucleus measurements. After exploratory analysis, cleaning and an 80:20 stratified split, hyperparameters were selected by 5-fold cross-validation (n_estimators = 200, max_depth = 10, max_features = 'log2'). On the held-out test set the model reached accuracy 95.61%, precision 100.00%, recall 88.10%, F1 93.67% and ROC-AUC 0.9942. The model produced 0 false positives but 5 false negatives, so the main weakness is missed malignancies, which is analysed together with the precision-recall trade-off.

1. Project Objective

The objective is to apply the Random Forest algorithm to a real-world classification problem and to demonstrate the full workflow: cleaning data, training an ensemble model, and evaluating it with professional metrics (confusion matrix, precision, recall, F1-score, ROC-AUC and feature importance). Every requirement of the project brief is mapped to a section of this report in the table below.

Brief requirement	Where it is addressed
Phase I: feature distributions, class imbalance, correlation heatmap	Section 3 (Figures 1, 2, 3)
Phase II: missing values, encoding, 80:20 split	Section 4
Feature scaling: justify use or skip	Section 4.4 and Section 7.2
Phase III: RandomForestClassifier, justify n_estimators and max_depth	Section 5 (Figures 4, 5)
Confusion-matrix heatmap	Section 6.1 (Figure 6)
Precision and recall trade-off discussion	Section 6.2 (Figure 7, Table 3)
F1-score	Section 6.2
ROC-AUC curve and AUC score	Section 6.3 (Figure 8)
Feature-importance bar chart	Section 6.4 (Figures 9, 10)
Conclusion: strengths, weaknesses, improvements	Section 8
Well-commented code	RF_Minor_Project.ipynb / rf_random_forest_project.py

2. Dataset and Problem Definition

The Breast Cancer Wisconsin (Diagnostic) dataset contains 569 patient records. Each record describes a digitised image of a fine-needle aspirate of a breast mass; ten characteristics of the cell nuclei (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension) are summarised by their **mean**, **standard error (se)** and **worst** (largest) value, giving 30 numeric features. The target is **diagnosis**: M (malignant) or B (benign). Malignant is treated as the positive class because failing to detect it is the costly error.

Why Random Forest suits this data: the problem is tabular and binary, features are numeric and heavily correlated, class boundaries are non-linear, the sample is small (so a variance-reducing ensemble helps), and the model needs no feature scaling and exposes feature importances.

3. Phase I: Exploratory Data Analysis

3.1 Class imbalance

The data contains 357 benign (62.7%) and 212 malignant (37.3%) cases, an imbalance ratio of 1.68:1. This is a mild imbalance, but it matters: a useless model that always answers 'benign' would already achieve 62.7% accuracy. Accuracy therefore cannot be the headline metric, and the analysis relies on precision, recall, F1 and ROC-AUC. A stratified split is used so both partitions keep the same class ratio.

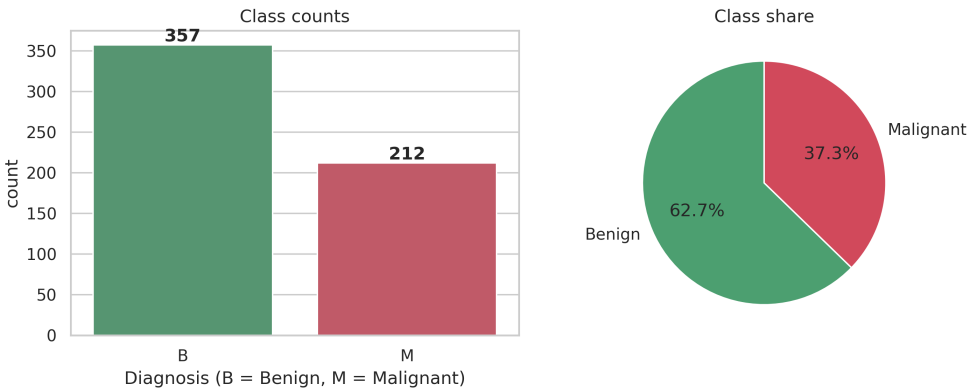


Figure 1. Class distribution of the target variable.

3.2 Feature distributions

Figure 2 shows the ten mean-features split by diagnosis. Size-related features (radius, perimeter, area) and shape-irregularity features (concavity, concave points) show clearly separated distributions, with malignant cells shifted to larger values; smoothness, symmetry and fractal dimension overlap heavily and are weaker individual predictors. Several features are right-skewed (e.g. area_mean skewness 1.65), which is harmless for tree-based models.

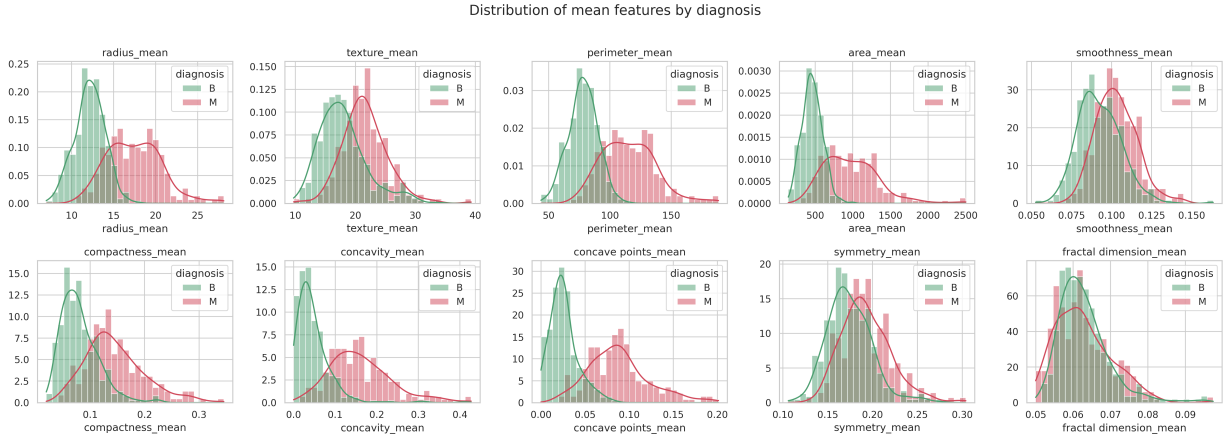


Figure 2. Distribution of mean features by diagnosis (histogram + KDE, normalised per class).

3.3 Correlation heatmap

Figure 3 is the correlation heatmap required by the brief. Two findings stand out. First, the features most correlated with malignancy are concave points_worst (r = 0.794), perimeter_worst, concave points_mean and radius_worst. Second, the features are strongly multicollinear: 21 feature pairs have |r| > 0.9 (radius, perimeter and area measure almost the same quantity, r up to 0.998). This does not hurt Random Forest's predictions, but it affects how feature importance must be interpreted (Section 6.4) and it is the reason why a linear model would need regularisation or feature selection.

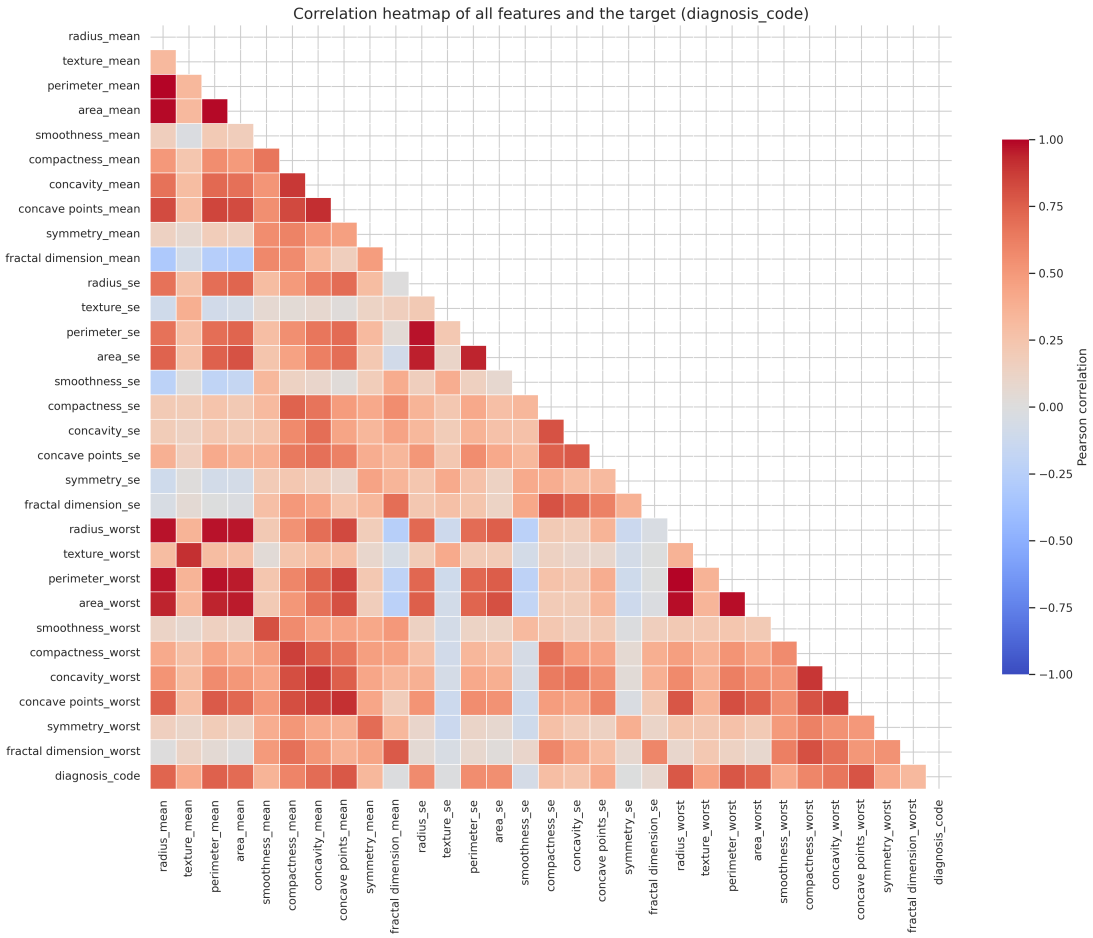


Figure 3. Correlation heatmap of all 30 features and the encoded target (lower triangle).

4. Phase II: Preprocessing and Data Cleaning

Step	Action taken	Reason / finding
Missing values	Checked every column. 569 null cells found, all in the empty export column 'Unnamed: 32'. Column dropped; remaining nulls = 0.	The column is 100% empty and carries no information. A median imputer fitted on the training split only is kept in the pipeline as a leakage-free safety net.
Irrelevant columns	Dropped 'id'.	A row identifier would let a tree memorise rows without learning anything general.
Encoding	Label-encoded the target: B = 0, M = 1.	Target is binary, so Label Encoding is enough (One-Hot would create two redundant columns). Fixing M = 1 makes malignant the positive class for precision/recall/F1.
Categorical features	None among the 30 features.	All features are numeric, so One-Hot encoding is not needed (the code checks for object columns and would handle them).
Train/test split	80:20 stratified split, random_state = 42: 455 train rows, 114 test rows.	Stratification keeps the malignant share at about 37% in both sets.
Feature scaling	Skipped.	See Section 4.4 and the experiment in Section 7.2.

Table 1. Preprocessing steps.

4.4 Feature-scaling decision

Random Forest chooses, at every node, a threshold on a single feature. Standardisation is a monotonic transformation of each feature, so it does not change the ordering of values and therefore does not change any split; the model is not distance-based. Scaling was therefore skipped. To verify this rather than assume it, a second forest with identical hyperparameters and seed was trained inside a StandardScaler pipeline: it produced **identical test predictions** (F1 0.9367, AUC 0.9942 in both cases).

5. Phase III: Model Development and Hyperparameter Justification

5.1 Baseline

A default forest (100 trees, unlimited depth) was trained first as a reference: test accuracy 97.37%, recall 92.86%, F1 96.30%, AUC 0.9929.

5.2 Choosing n_estimators and max_depth

n_estimators (number of trees). Averaging more trees reduces the variance of the ensemble, but with diminishing returns and linearly growing training and prediction cost. The goal is the smallest forest on the performance plateau. **max_depth** (maximum tree depth) controls the bias-variance trade-off of the individual trees: very shallow trees underfit, while unrestricted trees can memorise noise. Averaging makes forests tolerant of deep trees, but a depth cap still regularises and shrinks the model.

Both were chosen with a grid search using 5-fold stratified cross-validation on the training set only (the test set was never touched), over n_estimators in {50, 100, 200, 300}, max_depth in {3, 5, 7, 10, None} and max_features in {sqrt, log2} (40 configurations, 200 model fits). F1 was the selection metric because the brief names it as critical for imbalanced data; recall and ROC-AUC were recorded as well.

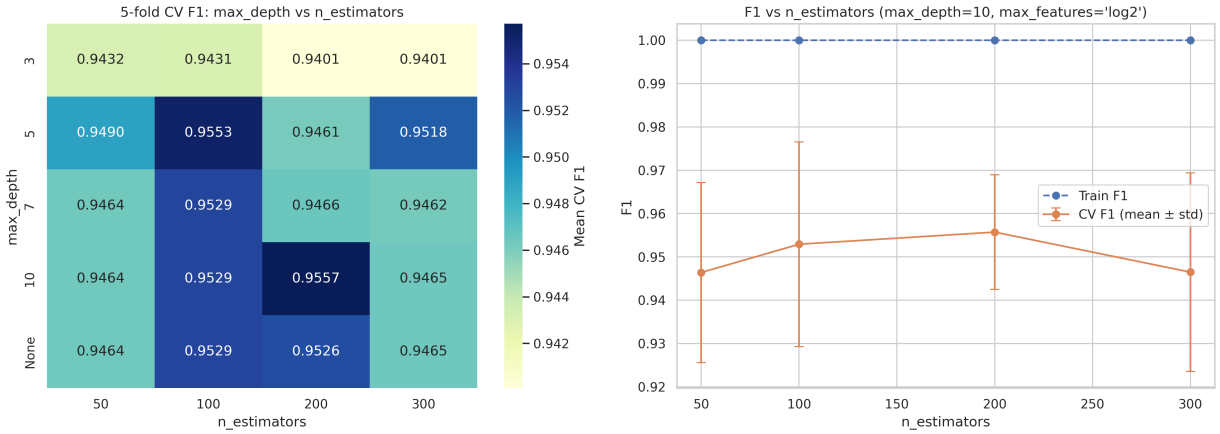


Figure 4. Left: mean CV F1 for every (max_depth, n_estimators) pair. Right: train vs cross-validated F1 as the number of trees grows (error bars = std over folds).

Result: n_estimators = 200, max_depth = 10, max_features = 'log2', mean CV F1 = 0.9557. Interpretation:

- **Depth:** depth 3 is clearly worse (CV F1 about 0.94), so very shallow trees underfit. From depth 5 upwards the score is essentially flat, i.e. the forest is not sensitive to this choice. Depth 10 is a sensible middle: deep enough to capture interactions, but capped to limit memorisation.
- **Trees:** performance does not keep rising with more trees; 100 and 200 trees perform best and 300 gives no gain, so extra trees only cost time. The OOB curve below agrees.
- **Honest caveat:** the whole grid spans only about 0.94 to 0.956 F1, and the fold-to-fold standard deviation is about 0.02. Only 9 of 40 settings are within 0.005 F1 of the best, but the differences among the top settings are within noise. The selected values are a reasonable, well-supported choice, not a uniquely 'correct' one.

- **Train vs CV gap:** train F1 is 1.0 for every forest size. This is normal for a random forest (each tree sees its own training rows) and is not by itself overfitting; the cross-validated score is the honest estimate.

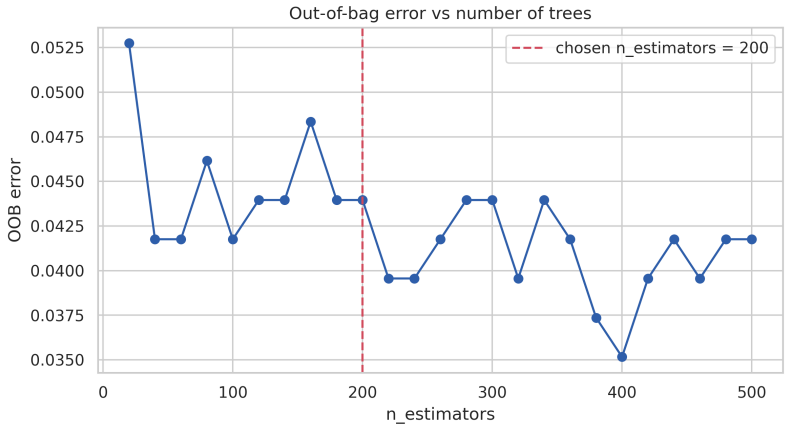


Figure 5. Out-of-bag error as the forest grows; the dashed line marks the chosen n_estimators.

The out-of-bag error drops from about 5.3% with 20 trees to about 4.2% by 40 trees and then fluctuates between roughly 3.5% and 4.8% with no clear downward trend (4.4% at the chosen 200 trees). Each step in the curve is a single training row (about 0.22 percentage points), so the wiggles, including the dip near 400 trees, are noise. The conclusion is that the forest stabilises early and that adding hundreds more trees is not worthwhile.

6. Evaluation on the Held-out Test Set

Metric	Baseline RF	Tuned RF (final)
Accuracy	97.37%	95.61%
Precision (malignant)	100.00%	100.00%
Recall (malignant)	92.86%	88.10%
F1-score (malignant)	96.30%	93.67%
ROC-AUC	0.9929	0.9942

Table 2. Test-set performance (114 patients).

6.1 Confusion matrix

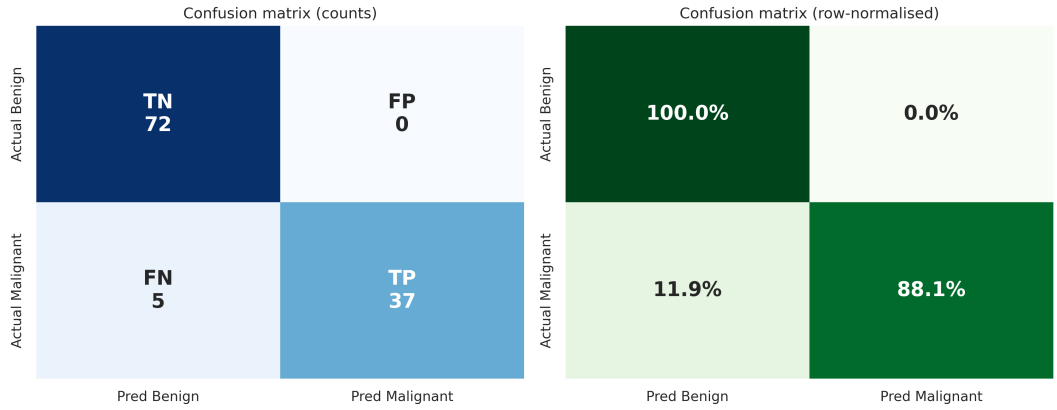


Figure 6. Confusion-matrix heatmap (counts and row-normalised percentages). Positive class = malignant.

Of the 72 benign patients, all 72 were correctly cleared (TN = 72, FP = 0). Of the 42 malignant patients, 37 were detected (TP) and **5 were missed (FN)**. In a screening context a false negative means a cancer that

is declared benign, which is far more dangerous than a false positive (an unnecessary follow-up test). The model's errors are all of the dangerous kind.

6.2 Precision, recall and F1

Precision = 100.00%. Every case the model flagged as malignant really was malignant: it is perfectly *exact*. **Recall = 88.10%.** The model found 37 of 42 malignant cases, so about 1 in 8 malignancies slipped through: it is not fully *complete*. **F1 = 93.67%**, the harmonic mean of the two, is pulled below the precision by the weaker recall, which is exactly why F1 is more informative than accuracy (95.61%) when classes are unbalanced. Put in terms of this dataset: *the model is very cautious about saying 'malignant', so it is never wrong when it does, but it misses too many actual cancers.*

The precision-recall trade-off. The 0.5 cut-off is a convention, not a law. Lowering the probability threshold makes the model flag more patients, which increases recall (fewer missed cancers) but lowers precision (more false alarms); raising it does the opposite. Table 3 and Figure 7 quantify this.

Threshold	Precision	Recall	F1	False positives	False negatives
0.2	0.8696	0.9524	0.9091	6	2
0.3	0.9512	0.9286	0.9398	2	3
0.4	0.9750	0.9286	0.9512	1	3
0.5	1.0000	0.8810	0.9367	0	5
0.6	1.0000	0.8810	0.9367	0	5
0.7	1.0000	0.8571	0.9231	0	6
0.8	1.0000	0.7857	0.8800	0	9

Table 3. Effect of the decision threshold on the test set (illustrative; a real threshold must be tuned on validation data, not on the test set).

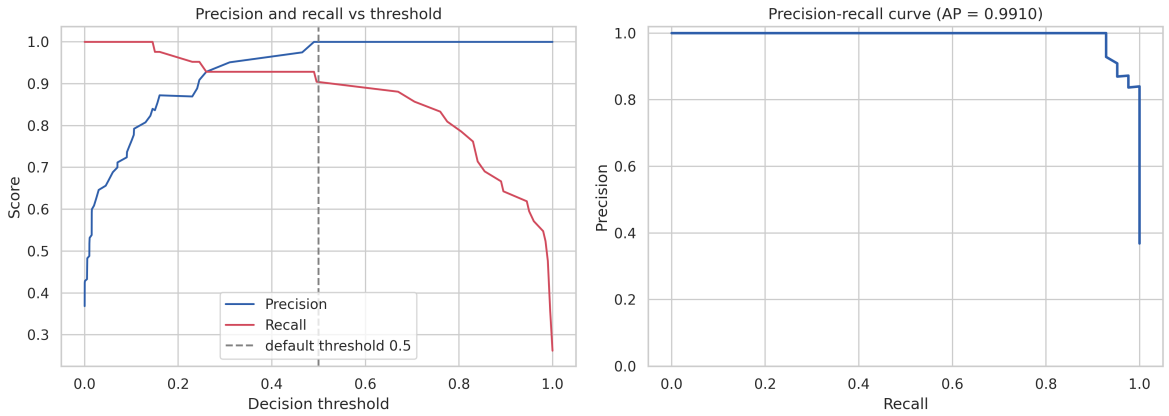


Figure 7. Left: precision and recall as the threshold varies. Right: precision-recall curve (average precision = 0.9910).

At a threshold of 0.4 the model trades one false positive for two fewer missed cancers (FN 5 to 3) and reaches the best F1 in the table (0.9512). At 0.2 recall rises to 95.2% but precision drops to 87.0% with 6 false alarms. Which point is 'best' is a clinical decision about the relative cost of the two errors; for cancer screening a threshold below 0.5 is easy to justify.

6.3 ROC curve and AUC

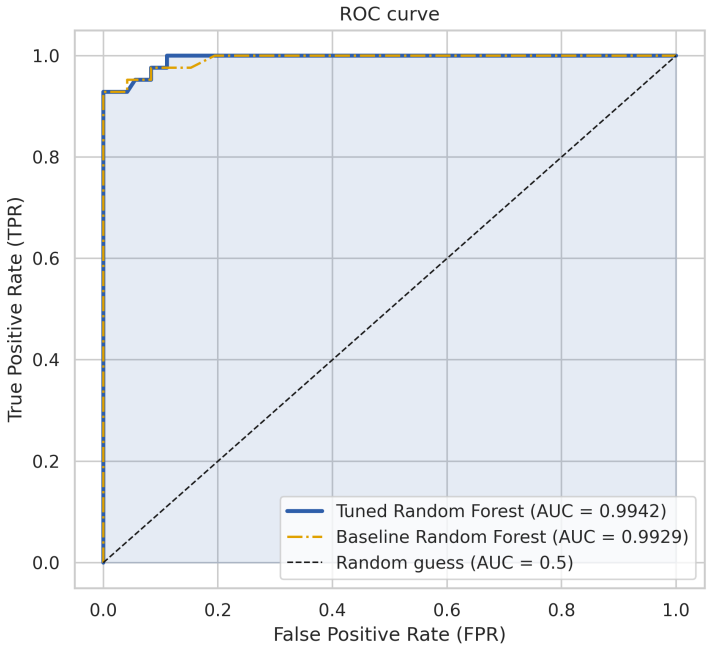


Figure 8. ROC curve of the tuned and baseline forests. Tuned AUC = 0.9942.

The ROC curve plots the true-positive rate against the false-positive rate over all thresholds. The curve hugs the top-left corner and the **AUC is 0.9942** (1.0 = perfect ranking, 0.5 = random). Interpretation: if one random malignant and one random benign patient are drawn, the model assigns the malignant one a higher malignancy probability about 99.4% of the time. The ranking ability is excellent even though recall at the 0.5 threshold is only 88.10%, which again shows that the weakness lies in the chosen cut-off rather than in the model's ability to separate the classes.

6.4 Feature importance

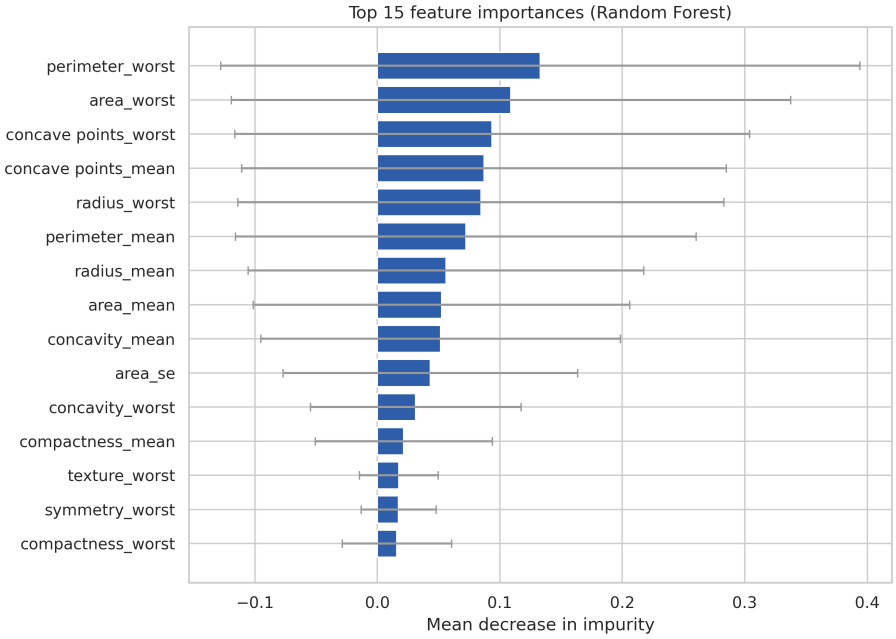


Figure 9. Top 15 impurity-based feature importances (error bars = std across trees).

The most important features are perimeter_worst (0.133), area_worst (0.109), concave points_worst (0.094) and concave points_mean (0.087). The top 5 features carry 50.8% of total importance and the top 10 carry 78.4%. The model relies mostly on the *worst* (largest) measurements of nucleus size and the concavity of the nucleus outline, which is clinically plausible: malignant nuclei tend to be larger and have more irregular, indented borders. The 'worst' values dominate because the single most abnormal cell in a

sample is more telling than the average.

Caution on interpretation. Because radius, perimeter and area are almost perfectly correlated (Section 3.3), the forest splits credit among them, so the importance of any one of them is understated and should be read as a group effect. Permutation importance (Figure 10, an independent check) shows this directly: shuffling any single feature barely hurts the test F1 because its correlated siblings compensate (largest drop only 0.0074). Only concave points_worst and texture_worst show a visible drop, and only two features appear in both top-10 lists. The two rankings agree on the broad picture (concavity and size features matter) but not on exact ordering.

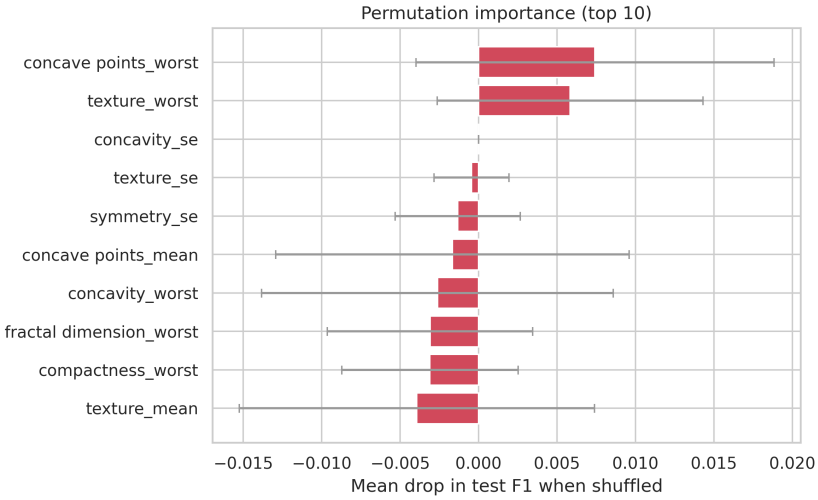


Figure 10. Permutation importance on the test set (drop in F1 when a feature is shuffled).

7. Robustness Checks and Model Comparison

7.1 Cross-validation on the full dataset

A single split of 114 test rows is noisy: one patient shifts recall by about 2.4 points (1 of 42 malignant cases) and accuracy by 0.9 points. The tuned forest was therefore also evaluated with 5-fold stratified cross-validation on all 569 rows:

Metric	Mean	Std over folds
accuracy	0.9596	0.0105
precision	0.9590	0.0314
recall	0.9341	0.0499
f1	0.9448	0.0157
roc_auc	0.9928	0.0039

Table 4. 5-fold cross-validated performance of the tuned Random Forest.

The cross-validated recall (93.4%) is higher than the single-split test recall (88.10%), and its fold-to-fold spread (+/- 5.0 points) is large. So the 5 misses on the test split are partly bad luck of that particular split; the realistic expectation is that the model misses roughly 5 to 10% of malignancies at the default threshold. Note also that the tuned forest scored slightly *below* the default forest on this test split (95.61% vs 97.37% accuracy, a difference of two patients) even though it scored marginally higher in cross-validation (CV F1 0.9557 for the selected setting vs 0.9529 for the closest default-like grid setting); with only 114 test rows this is within noise and illustrates why tuning must be judged by cross-validation rather than by one split.

7.2 Scaling experiment

As reported in Section 4.4, a StandardScaler + Random Forest pipeline produced exactly the same test predictions as the unscaled forest, confirming that scaling is unnecessary.

7.3 Comparison with Gradient Boosting

Metric	Baseline RF	Tuned RF	Gradient Boosting (default)
Accuracy	0.9737	0.9561	0.9649
Precision	1.0000	1.0000	1.0000
Recall	0.9286	0.8810	0.9048
F1	0.9630	0.9367	0.9500
ROC-AUC	0.9929	0.9942	0.9947

Table 5. Untuned Gradient Boosting compared with the Random Forests on the same test split.

An untuned Gradient Boosting classifier is competitive (F1 0.9500, AUC 0.9947) and slightly better than the tuned forest on this split, but the differences amount to one or two patients and are not statistically meaningful. It does suggest that boosting plus tuning is a worthwhile next experiment, as proposed in the conclusion.

8. Conclusion

A Random Forest with 200 trees and a maximum depth of 10 classifies breast-tumour diagnoses with 95.61% test accuracy, perfect precision, 88.10% recall, F1 of 93.67% and an ROC-AUC of 0.9942.

Strengths

- Excellent class separation (AUC 0.994) and perfect precision at the default threshold: no benign patient was wrongly flagged on the test split.
- Minimal preprocessing: no scaling (verified), no sensitivity to skewed or correlated features, no encoding needed beyond the target.
- Stable behaviour: the score is almost flat across a wide range of n_estimators and max_depth, and cross-validation (F1 0.945, AUC 0.993) confirms the result is not a lucky split.
- Interpretable output: feature importance points to size and concavity measurements of the nuclei, which matches medical intuition.

Weaknesses

- **Missed malignancies:** recall is 88.10%; 5 of 42 cancers were classified benign. In screening, this is the costliest type of error, and the 0.5 threshold does not reflect that cost.
- **Small test set:** with 114 rows, each patient moves recall by about 2.4 points, so the confidence intervals are wide and small model differences cannot be trusted.
- **Correlated features:** importance is shared among radius/perimeter/area, so single-feature importance values are unreliable (permutation importance shows near-zero values for this reason).
- **Generalisation:** the data comes from one source and 569 samples; performance on other hospitals, scanners or populations is unknown. The forest is also less transparent than a single decision tree or logistic regression.

Potential improvements

- Tune the decision threshold (or use class_weight) on a validation set to raise recall; Table 3 shows that a threshold near 0.3-0.4 would have cut false negatives from 5 to 3 at the cost of 1-2 false positives.

- Try Gradient Boosting, XGBoost or LightGBM with proper hyperparameter search, and compare with logistic regression and SVM baselines.
- Use nested cross-validation for an unbiased tuned estimate, calibrate the probabilities (CalibratedClassifierCV) and add SHAP values for per-patient explanations.
- Reduce redundancy with feature selection or PCA on the correlated size features, and validate on an external dataset before any real-world use.

This project is an educational exercise and not a diagnostic tool.

9. Reproducibility

Files: **RF_Minor_Project.ipynb** (commented notebook), **rf_random_forest_project.py** (same code as a script), **plots/** (10 PNG figures at 300 dpi), **results.json** (all reported numbers). Requirements: Python 3, pandas, numpy, scikit-learn, matplotlib, seaborn. A fixed seed (random_state = 42) is used everywhere. Place Kaggle's **data.csv** next to the notebook; if absent, the code loads the identical dataset bundled with scikit-learn and rebuilds Kaggle's column layout.